

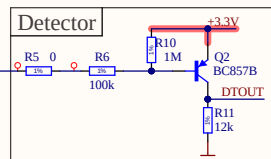
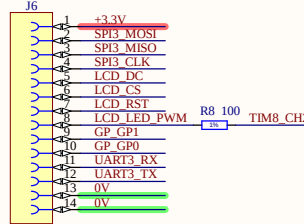
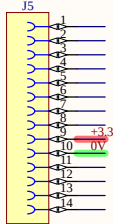
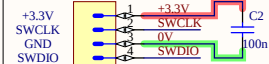
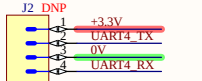
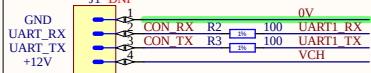
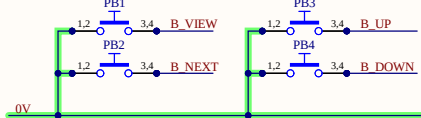
A

B

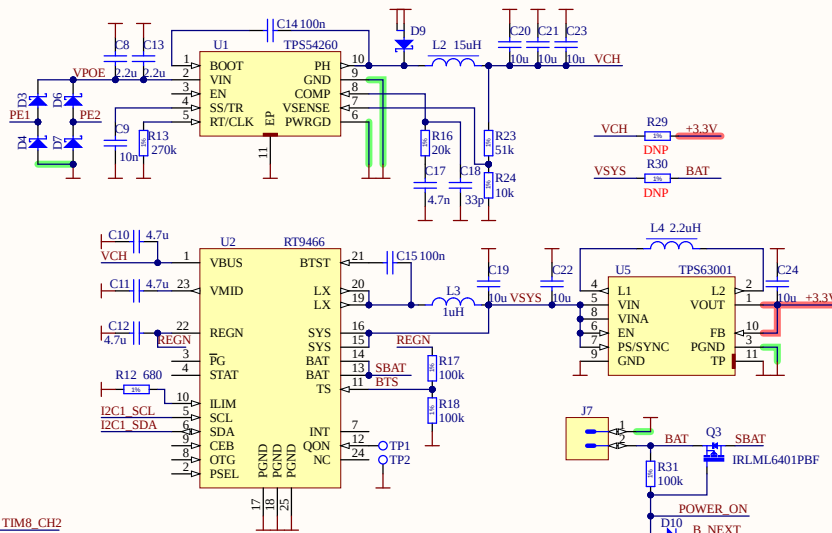
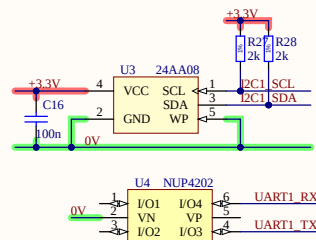
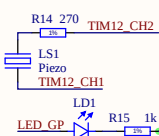
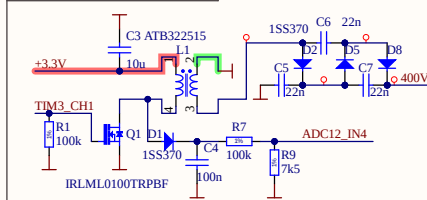
C

D

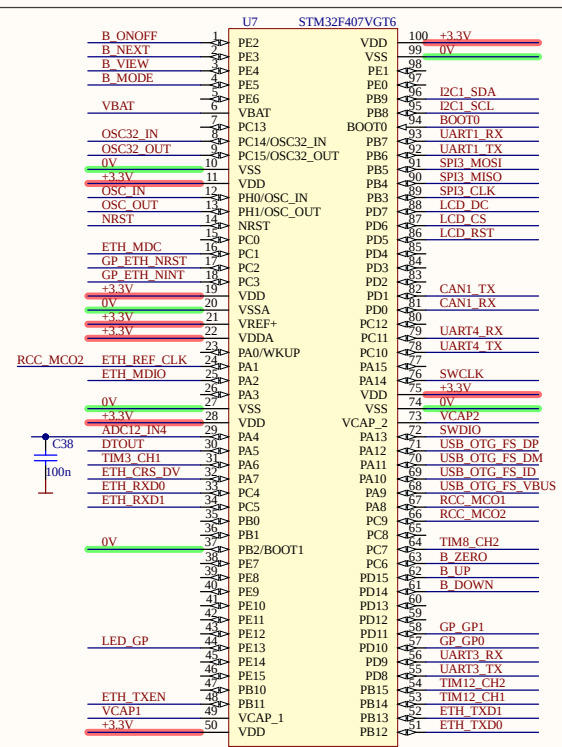
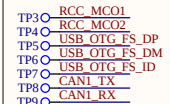
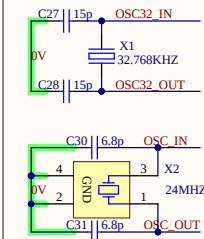
KEY BOARD



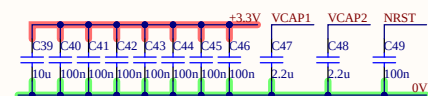
High Voltage DC/DC



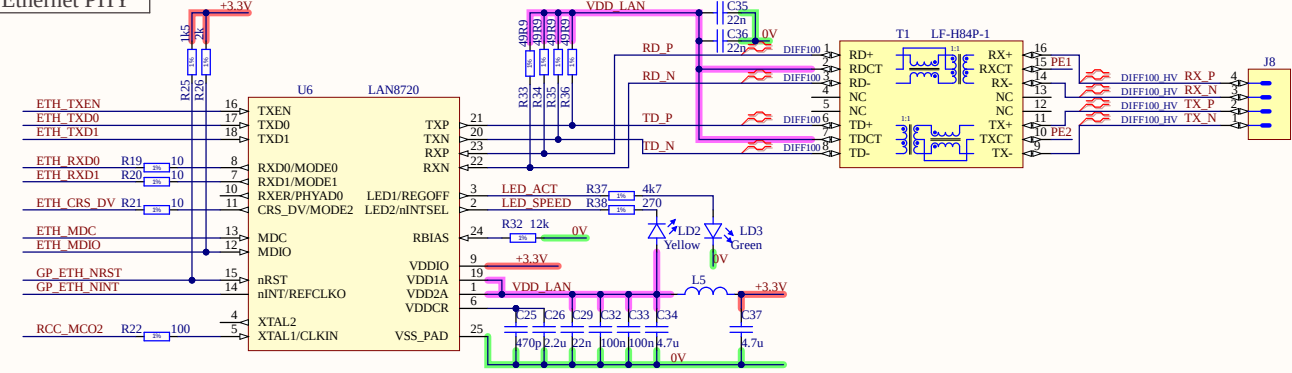
MCU Signals



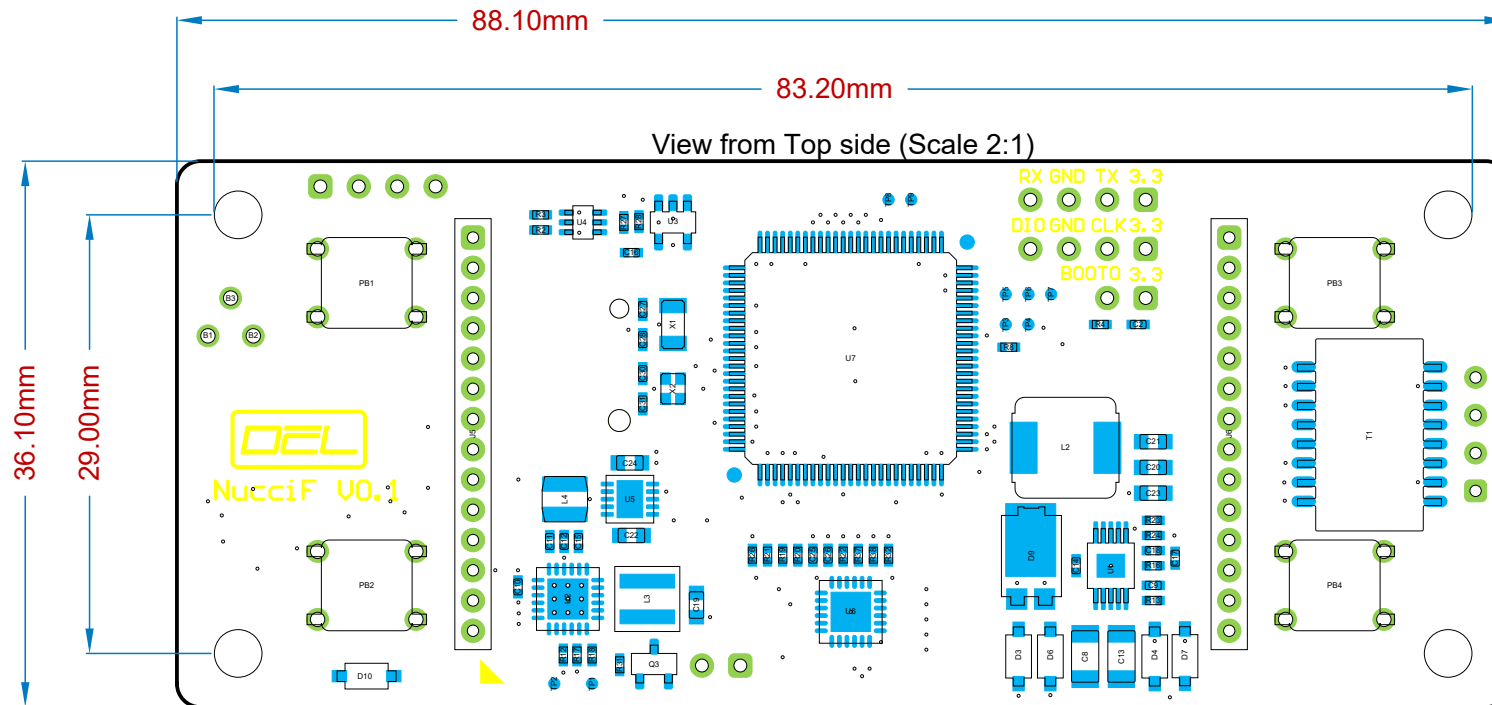
Issue / ToDo:
- need connect POWER_ON to PA3



Ethernet PHY



Designator	Comment	Footprint	Part Number	Part Description	Quantity
B1, B2, B3	V, P, G	c150h100	PLATED_HOLE1.0_PAD1.5	Plated Through Hole: Hole Dia.=1.0mm Pad Dia.=1.5mm	3
C1, C2, C4, C16, C32, C33, C38, C40, C41, C42, C43, C44, C45, C46, C49	100n	CAPC1005X55N	CC0402_100NF_50V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	15
C3, C19, C20, C21, C22, C23, C24, C39	10u	CAPC1608X87N	CC0603_10UF_10V_20%_X5R	SMD Multilayer Chip Ceramic Capacitor	8
C5, C6, C7	22n	CAPC2012X135N	CC0805_22NF_500V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	3
C8, C13	2.2u	CAPC3216X140N	CC1206_2.2UF_100V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	2
C9	10n	CAPC1005X55N	CC0402_10NF_16V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	1
C10, C11, C12	4.7u	CAPC1005X55N	CC0402_4.7UF_10V_20%_X5R	SMD Multilayer Chip Ceramic Capacitor	3
C14, C15	100n	CAPC1005X55N	CC0402_100NF_16V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	2
C17	4.7n	CAPC1005X55N	CC0402_4.7NF_16V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	1
C18	33p	CAPC1005X55N	CC0402_33PF_50V_5%_NPO	SMD Multilayer Chip Ceramic Capacitor	1
C25	470p	CAPC1005X55N	CC0402_470PF_50V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	1
C26, C47, C48	2.2u	CAPC1005X55N	CC0402_2.2UF_6.3V_20%_X5R	SMD Multilayer Chip Ceramic Capacitor	3
C27, C28	15p	CAPC1005X55N	CC0402_15PF_50V_5%_NPO	SMD Multilayer Chip Ceramic Capacitor	2
C29, C35, C36	22n	CAPC1005X55N	CC0402_22NF_16V_10%_X7R	SMD Multilayer Chip Ceramic Capacitor	3
C30, C31	6.8p	CAPC1005X55N	CC0402_6.8PF_50V_0.5PF_NPO	SMD Multilayer Chip Ceramic Capacitor	2
C34, C37	4.7u	CAPC1608X87N	CC0603_4.7UF_10V_10%_X5R	SMD Multilayer Chip Ceramic Capacitor	2
D1, D2, D5, D8	1SS370	SOT65P210X100-3N	1SS370	High Voltage, High Speed Switching Applications	4
D3, D4, D6, D7	MBR0540T1G	SOD3716X135N	MBR0540T1G	Schottky Power Rectifier	4
D9	SBRT15U100SP5	SBRT15U100SP5	SBRT15U100SP5	Schottky Rectifier	1
D10	1N4148W	SOD3715X145N	1N4148W	Small Signal Diode	1
J1, J2, J3	Header 4	3M_2304-6111-TG	3M_2304-6111-TG	4 Contacts - Vertical Contacts Straight Strip Header Pitch 2.54	3
J4, J7	Header 2	3M_2302-6111-TG	3M_2302-6111-TG	2 Contacts - Horizontal Contacts Straight Strip Header Pitch 2.54	2
J5, J6	Socket 14	3M_955114-6102-AR	3M_955114-6102-AR	14 Contacts - 2mm x 2 mm Straight, Boardmount Socket, 950 Serie	2
J8	SPOX	MOLEX_22-03-5045	MOLEX_22-03-5045	2.50mm Pitch SPOX Wire-to-Board Header, Vertical, Shrouded, 4 Circuits, Tin (Sn) Plating	1
L1	ATB322515	XFMR_ATB322515-0110-T000	XFMR_TDK_ATB322515-0110-T000	1:10 Step-up Transformers	1
L2	15uH	IND_VISHAY_IHLP-2525C2	IND_15UH_20%_HP10630	SMT Inductors SIMID Series	1
L3	1uH	IND_COILCRAFT_XAL4020	IND_1UH_20%_COILCRAFT_XAL4020-102ME	Shielded SMD Power Inductors	1
L4	2.2uH	VLS3010CX-1	IND_2.2UH_20%_TDK_VLS3010CX-2R2M-1	SMT Power Inductors	1
L5	2.2uH	INDC2520X100N	IND1008_2.2UH_20%_TDK_MLP2520S2R2MT0S1	SMT Inductors	1
LD1	Blue	LED1608X80N_BLUE	LED_OF-SMD1608B	Surface Mount Red LED (0603 Case)	1
LD2	Yellow	LED1608X80N_YELLOW	LED_OF-SMD1608Y	Surface Mount Red LED (0603 Case)	1
LD3	Green	LED1608X80N_GREEN	LED_OF-SMD1608G	Surface Mount Green LED (0603 Case)	1
LS1	Piezo	HPS13C	HPS13C	Piezo sound generator	1
PB1, PB2, PB3, PB4	Push Button	PB_E-SWITCH_TL1105HF100Q	PB_E-SWITCH_TL1105HF100Q	12VDC 50mA SPST Tactile Push-Button Switch	4
Q1		IRLML0100TRPBF	IRLML0100TRPBF	100V 1.6A N-Channel HEXFET* Power MOSFET	1
Q2		BC857B	SOT95P230X110-3N	45V, 100mA PNP general-purpose transistors	1
Q3		IRLML6401PBF	SOT95P237X112-3N	-12V -4.3A P-Channel HEXFET* Power MOSFET	1
R1, R7, R17, R18, R31	100k	RESC1005X40N	R0402_100K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	5
R2, R3, R8, R22	100	RESC1005X40N	R0402_100R_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	4
R4, R15	1k	RESC1005X40N	R0402_1K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	2
R5	0	RESC2012X60N	R0805_0R_JUMPER	2A (0.05R Max DC Resistance) Zero Ohm Jumper	1
R6	100k	RESC2012X60N	R0805_100K_1%_0.125W_100PPM	General Purpose Thick Film Chip Resistor	1
R9	7k5	RESC1005X40N	R0402_7K5_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R10	1M	RESC1005X40N	R0402_1M_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R11, R32	12k	RESC1005X40N	R0402_12K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	2
R12	680	RESC1005X40N	R0402_680R_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R13	270k	RESC1005X40N	R0402_270K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R14, R38	270	RESC1005X40N	R0402_270R_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	2
R16	20k	RESC1005X40N	R0402_20K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R19, R20, R21	10	RESC1005X40N	R0402_10R_1%_0.0625W_200PPM	General Purpose Thick Film Chip Resistor	3
R23	51k	RESC1005X40N	R0402_51K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R24	10k	RESC1005X40N	R0402_10K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R25	1k5	RESC1005X40N	R0402_1K5_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
R26, R27, R28	2k	RESC1005X40N	R0402_2K_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	3
R29, R30	0	RESC1005X40N	R0402_0R_JUMPER	1A (0.05R Max DC Resistance) Zero Ohm Jumper	2
R33, R34, R35, R36	49R9	RESC1005X40N	R0402_49R9_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	4
R37	4k7	RESC1005X40N	R0402_4K7_1%_0.0625W_100PPM	General Purpose Thick Film Chip Resistor	1
SK1	BH500	EEMB_BH500	EEMB_BH500	Battery holder for CR and LIR series L216 to L220, L225	1
T1	LF-H84P-1	XFMR_LF-H84P	PE-68678	10/100BASE-Tx PC Card LAN Magnetics Module	1
TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9	Test Point, GND	TP_SMD_800	TP_SMD_0.8MM	0.8mm round SMD Pad	9
U1	TPS54260	SOP50P490X110-11N-R157X188	TPS54260DGQ	3.5-V to 60-V Input, 2.5-A, Step-Down Converter	1
U2	RT9466	QFN50P400X400X100-25N-S270	RT9466	Linear Li-Ion Charger with LDO	1
U3	24AA08	SOT95P280X145-5N	24AA08T-I/OT	8K Serial EEPROM	1
U4	NUP4202	SOT65P220X110-6N	NUP4202W1	ESD Protection Diode, Low Clamping Voltage	1
U5	TPS63001	SON50P300X300X100-11N-R175X250	TPS63001	Single Inductor Buck-Boost Converter With 1.8-A Switches	1
U6	LAN8720	QFN50P400X400X100-25N-S265	LAN8720A-CP-TR	Small Footprint RMII 10/100 Ethernet Transceiver with HP Auto-MDIX Support	1
U7	STM32F407VGT6	QFP50P1600X1600X160-100N	STM32F407VGT6	ARM Cortex-M4 32b MCU+FPU, 210DMIPS, up to 1MB Flash/192+4KB RAM, USB OTG HS/FS, Ethernet, 17 TIMs, 3 ADCs, 15 comm. interfaces & camera	1
X1	32.768KHZ	OSCCC150X320X80-2N	32.768KHz_KX-327NHT	Quartz Crystal ±20ppm	1
X2	24MHZ	OSCCC160X200X65-4N	24MHZ_XRCGB24M000F3M00R0	± 30ppm at (25±3°C), -30°C to +85°C, 6pF, Ceramic SMD Crystal, Type HCR2016	1



View from Bottom side (Scale 2:1)

